Paper 3 – Suggested Solutions

1 **(a)** $t = \dfrac{v-u}{a}$

Sub into the 2nd equation

$$s = \frac{1}{2}(u+v)\left(\frac{v-u}{a}\right) = \frac{(v+u)(v-u)}{2a}$$

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$

(b) Displacement required $= 3.0 + \dfrac{0.23}{2} - 2.1 = 1.015$ m

Minimum speed of release required means that velocity at 1.015 m is zero.

$$v^2 = u^2 + 2as$$

$$0 = u^2 + 2(-9.81)(1.015)$$

$$u = 4.46 \text{ m s}^{-1}$$

OR,

using conservation of energy

Loss in kinetic energy = Gain in GPE

$$\frac{1}{2}mu_y^2 - 0 = mgh_f - mgh_i$$

$$\frac{1}{2}u_y^2 = g(h_f - h_i) = gs_y$$

$$u_y^2 = \sqrt{2(9.81)(1.015)}$$

$$u_y = 4.46 \text{ m s}^{-1}$$

(c) **(i)** Horizontally

$$s_x = u_x t = ut\cos\theta$$

$$6.7 = u(1.3)\cos 50^\circ$$

$$u = \frac{6.7}{1.3\cos 50^\circ} = 8.02 \text{ m s}^{-1}$$

(ii) Vertically

$$s_y = u_y t + \frac{1}{2}a_y t^2 = ut\sin 50^\circ + \frac{1}{2}(-9.81)(1.3)^2$$

$$= 8.0\sin 50^\circ \times 1.3 + \frac{1}{2}(-9.81)(1.3)^2$$

$$= -0.3226 \text{ m}$$

height = 2.5 – 0.3226 = 2.18 m

2 **(a)** Gravitational potential (or potential energy) at infinity is zero